

Inequations

Inequations, also known as inequalities, represent relationships between two quantities where they are not necessarily equal. They use symbols such as $>$, $<$, $\geq$, and $\leq$ to indicate whether one quantity is greater than, less than, greater than or equal to, or less than or equal to the other. Inequations can involve linear expressions, indicating a range of possible solutions, or more complex forms like absolute value inequalities, requiring consideration of multiple cases to capture all possible solutions. When solving inequalities, the focus is on isolating the variable while maintaining the relationship denoted by the inequality symbol, with special care taken when multiplying or dividing by negative numbers. Inequations are key in algebra, calculus, and optimization, providing a way to describe conditions where equality does not necessarily hold.

Example 7: Solve $7x^4 > 2x^3 + 9x^2$

To solve the inequality $7x^4 > 2x^3 + 9x^2$, let's first move all terms to one side to obtain a unified expression, and then analyze the resulting polynomial to find the critical points (roots) and intervals where the inequality holds true.

Move all terms to one side to make it easier to find the roots:

$$7x^4 - 2x^3 - 9x^2 > 0.$$

One way to solve this type of inequality is to look for common factors or apply algebraic techniques to identify the critical points:

In this case, the common factor is x^2 :

$$x^2(7x^2 - 2x - 9).$$

To determine where the polynomial changes sign, set each factor to zero:

$$x^2 = 0 \Rightarrow x = 0.$$

For $7x^2 - 2x - 9$ use the quadratic formula to find roots:

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \times 7 \times -9}}{2 \times 7}.$$

Simplifying, we have:

$$x = \frac{2 \pm \sqrt{4 + 252}}{14} \Rightarrow x = \frac{2 \pm \sqrt{256}}{14} \Rightarrow x = \frac{2 \pm 16}{14}.$$

This yields two roots: $x = \frac{18}{14} = \frac{9}{7}$ and $x = \frac{-14}{14} = -1$.

These roots indicate where the expression changes sign, allowing analysis of the intervals to determine where the inequality is greater than zero.

Given the roots $x = 0$, $x = \frac{9}{7}$, and $x = -1$, let's consider the intervals created by these roots:

When $x < -1$, the expression results in a positive value.

When $-1 < x < 0$, the expression results in a negative value.

When $0 < x < \frac{9}{7}$, this results in a negative value.

When $x > \frac{9}{7}$, this results in a positive value

$-\infty$	...	-1	...	0	...	$\frac{9}{7}$	...	$+\infty$
	>0	$= 0$	<0	$= 0$	<0	$= 0$		>0
	$\uparrow$							$\uparrow$

Thus, the solution for $7x^4 > 2x^3 + 9x^2$ is $x \in (-\infty, -1] \cup [\frac{9}{7}, +\infty)$.

Example 8: Solve $2x^6 - 2 > -3x^3$

To solve the inequality $2x^6 - 2 > -3x^3$, first move all terms to one side of the inequality:

$$2x^6 + 3x^3 - 2 > 0.$$

To find the critical points where the expression equals zero, set the polynomial to zero and solve for x :

$$2x^6 + 3x^3 - 2 = 0.$$

We can consider x^3 as a common term and rewrite this expression as a quadratic equation in terms of x^3 :

$$2(x^3)^2 + 3x^3 - 2 = 0.$$

To find the roots, use the quadratic formula:

$$x^3 = \frac{-3 \pm \sqrt{3^2 - 4 \times 2 \times -2}}{2 \times 2},$$

$$x^3 = \frac{-3 \pm \sqrt{25}}{4} \Rightarrow x^3 = \frac{-3 \pm 5}{4}.$$

This yields the roots:

$$1. x^3 = \frac{2}{4} = \frac{1}{2} \Rightarrow x = \sqrt[3]{\frac{1}{2}}. \Rightarrow x = \frac{1}{\sqrt[3]{2}}$$

$$2. x^3 = \frac{-8}{4} = -2. \Rightarrow x = -\sqrt[3]{2}$$

Given these critical points, let's examine the intervals to determine where the original expression is greater than zero:

When $x < -\sqrt[3]{2}$, it results in a positive value, indicating this interval satisfies the inequality.

When $-\sqrt[3]{2} < x < \frac{1}{\sqrt[3]{2}}$ the expression results in a negative value, indicating this interval does not satisfy the inequality.

When $x > \frac{1}{\sqrt[3]{2}}$ the result is positive, indicating this interval satisfies the inequality.

$-\infty$	...	$-\sqrt[3]{2}$	...	$\frac{1}{\sqrt[3]{2}}$	...	$+\infty$
>0		$= 0$	<0	$= 0$	>0	
$\uparrow$					$\uparrow$	

Thus, the solution for $2x^6 - 2 > -3x^3$ is:

$$x \in (-\infty, -\sqrt[3]{2} \cup \frac{1}{\sqrt[3]{2}}, +\infty).$$